Higgs Precision and BSM Benchmarks at LHμC

Charged-current $h \to b\bar{b}$, κ-framework fits, RPV squarks, and color-octet muons

April 28, 2026

Abstract

These notes collect the physics discussion behind four presentation topics: (i) the precision Higgs case study $p\mu^+ \to j\bar{\nu}_\mu h$, $h \to b\bar{b}$; (ii) the broader Higgs-coupling reach in a κ-framework global-fit picture; (iii) why LHμC is also a distinctive BSM machine; and (iv) representative BSM benchmarks, namely R-parity-violating squarks and color-octet muons. The emphasis is pedagogical: what the slides are claiming, how the numbers are obtained, what the plots mean, and what caveats should be stated during a conference talk.

1 Executive summary

The short physics story

LHμC is not simply an energy upgrade of the LHeC. It opens a multi-TeV lepton–hadron regime where weak-boson-fusion Higgs production, high-Q^2 electroweak measurements, photon-induced reactions, top production, and direct $\mu q/\mu g$ BSM resonances become quantitatively competitive.

The four target slides should be understood as one connected narrative:

1. **Concrete Higgs benchmark:** charged-current VBF production, $p\mu^+ \to j\bar{\nu}_\mu h$, followed by $h \to b\bar{b}$, can be translated into a sensitivity on g_{hbb}.
2. **Global Higgs fit:** beyond a single channel, the strongest visible improvement is in vector modifiers, especially κ_W and κ_Z, because Higgs production at LHμC is dominated by weak-boson fusion.
3. **BSM logic:** LHμC is special because it directly probes μq and μg initial states at multi-TeV energy.
4. **BSM benchmarks:** RPV squarks test the μq portal, while color-octet muons test the μg portal.

2 Benchmark setup and notation

The current presentation uses the following working points:

Table 1: Nominal LHμC benchmark points used in the slides.

Scenario	E_μ	E_p	$\sqrt{s_{\mu p}}$ and luminosity	
LHμC baseline	500 GeV	7 TeV	$\sqrt{s} \simeq 3.7$ TeV,	$L_{\rm int} = 250$ fb^{-1}
LHμC upgrade	1 TeV	7 TeV	$\sqrt{s} \simeq 5.3$ TeV,	$L_{\rm int} = 1000$ fb^{-1}
Possible higher stage	2 TeV	7 TeV	$\sqrt{s} \simeq 7.5$ TeV,	$L_{\rm int} = 1000$ fb^{-1}

Important consistency issue

The $h \to b\bar{b}$ table in the slide gives $\Delta\sigma/\sigma = 0.69\%$ for the 1 TeV LHμC benchmark. Using the table cross sections and efficiencies, this value corresponds more closely to $L_{\rm int} = 500$ fb^{-1}, not 1000 fb^{-1}. If the slide keeps the global benchmark $L_{\rm int} = 1000$ fb^{-1}, the rate uncertainty should be closer to 0.49%, under the same assumptions. This point should be fixed before the talk.

3 Precision Higgs case study: $h \to b\bar{b}$

3.1 Why charged-current VBF dominates

At a lepton–hadron collider, Higgs production is mainly electroweak vector-boson fusion:

$$\text{CC VBF:} \quad p\mu^+ \to j\,\bar{\nu}_\mu\,h, \tag{1}$$
$$\text{NC VBF:} \quad p\mu^+ \to j\,\mu^+\,h. \tag{2}$$

The charged-current process is larger because it is driven by W-exchange. In the current benchmarks, the slides quote approximately

$$\sigma_{\rm CC}(500\ \text{GeV}) \simeq 438\ \text{fb}, \qquad \sigma_{\rm NC}(500\ \text{GeV}) \simeq 135\ \text{fb}, \tag{3}$$
$$\sigma_{\rm CC}(1\ \text{TeV}) \simeq 731\ \text{fb}, \qquad \sigma_{\rm NC}(1\ \text{TeV}) \simeq 221\ \text{fb}. \tag{4}$$

This already gives large Higgs statistics:

$$N_{\rm CC}(500\ \text{GeV}) \simeq 438\ \text{fb} \times 250\ \text{fb}^{-1} \simeq 1.10 \times 10^5, \tag{5}$$
$$N_{\rm CC}(1\ \text{TeV}) \simeq 731\ \text{fb} \times 1000\ \text{fb}^{-1} \simeq 7.31 \times 10^5. \tag{6}$$

Why the muon beam matters

The physics is analogous to the LHeC, but a muon beam allows much higher lepton energy because synchrotron radiation is much weaker for muons than electrons. This raises $\sqrt{s_{\mu p}}$, increases the hard-scattering phase space, and enhances Higgs, top, electroweak, and BSM rates.

3.2 Why $h \to b\bar{b}$ is the natural benchmark

The decay $h \to b\bar{b}$ is the largest Standard Model Higgs decay mode and directly probes the bottom Yukawa coupling. It is experimentally nontrivial because the final state contains hadronic jets and backgrounds from $Z \to b\bar{b}$ and multijet production. However, the lepton–hadron environment is cleaner than pp collisions and the topology is more constrained.

The benchmark signal is

$$p\mu^+ \to j\,\bar{\nu}_\mu\,h, \qquad h \to b\bar{b}. \tag{7}$$

The dominant backgrounds are

$$p\mu^+ \to j\,\bar{\nu}_\mu\,Z, \qquad Z \to b\bar{b}, \tag{8}$$
$$p\mu^+ \to \bar{\nu}_\mu\,j\,j\,j. \tag{9}$$

3.3 Analysis chain

The benchmark analysis uses

- MG5_aMC@NLO for event generation;
- Pythia6 for showering and hadronization;
- FastJet for jet clustering;
- Delphes with an LHeC-like asymmetric detector card;
- 75% b-tagging efficiency;
- exactly two reconstructed b-jets;
- the Higgs mass-window cut

$$|m_{bb} - m_h| < 25\ \text{GeV}. \tag{10}$$

3.4 Rate and coupling interpretation

The measured final-state rate is

$$\sigma(p\mu^+ \to j\bar{\nu}_\mu b\bar{b}) = \sigma(p\mu^+ \to j\bar{\nu}_\mu h)\,\mathrm{BR}(h \to b\bar{b}). \tag{11}$$

Since charged-current VBF scales as g^2_{hWW}, while

$$\mathrm{BR}(h \to b\bar{b}) \propto \frac{g^2_{hbb}}{\Gamma_h},$$

the final-state rate scales as

$$\sigma(p\mu^+ \to j\bar{\nu}_\mu b\bar{b}) \propto \frac{g^2_{hWW}\, g^2_{hbb}}{\Gamma_h}. \tag{12}$$

Thus the hbb channel does not measure g_{hbb} alone. It measures a product involving the production coupling, the decay coupling, and the total Higgs width.

The event yields are

$$N_s = L_{\mathrm{int}}\, \sigma_{\mathrm{sig}}\, \epsilon_{\mathrm{sig}}, \tag{13}$$

$$N_b = L_{\mathrm{int}} \left[\sigma_{\mathrm{bgd-Z}}\epsilon_{\mathrm{bgd-Z}} + \sigma_{\mathrm{bgd-mj}}\epsilon_{\mathrm{bgd-mj}}\right]. \tag{14}$$

The statistical rate uncertainty is

$$\frac{\Delta\sigma}{\sigma} = \frac{\sqrt{N_s + N_b}}{N_s}. \tag{15}$$

To extract g_{hbb}, external information on g^2_{hWW}/Γ_h is used:

$$\frac{\Delta g_{hbb}}{g_{hbb}} = \frac{1}{2}\sqrt{\left(\frac{\Delta\sigma}{\sigma}\right)^2 + \left[\frac{\Delta(g^2_{hWW}/\Gamma_h)}{g^2_{hWW}/\Gamma_h}\right]^2}. \tag{16}$$

The factor 1/2 appears because the rate depends on g^2_{hbb}, not linearly on g_{hbb}.

3.5 Numerical check

For the 500 GeV benchmark, using

$$L_{\mathrm{int}} = 250\ \mathrm{fb}^{-1} = 250000\ \mathrm{pb}^{-1},$$

$$\sigma_{\mathrm{sig}} = 0.24\ \mathrm{pb}, \quad \epsilon_{\mathrm{sig}} = 0.22,$$

$$\sigma_{\mathrm{bgd-Z}} = 0.30\ \mathrm{pb}, \quad \epsilon_{\mathrm{bgd-Z}} = 0.025,$$

$$\sigma_{\mathrm{bgd-mj}} = 357.9\ \mathrm{pb}, \quad \epsilon_{\mathrm{bgd-mj}} = 9.7 \times 10^{-5},$$

one obtains

$$N_s \simeq 250000 \times 0.24 \times 0.22 \simeq 13200, \tag{17}$$

$$N_{b,Z} \simeq 250000 \times 0.30 \times 0.025 \simeq 1875, \tag{18}$$

$$N_{b,\mathrm{mj}} \simeq 250000 \times 357.9 \times 9.7 \times 10^{-5} \simeq 8680. \tag{19}$$

Then

$$\frac{\Delta\sigma}{\sigma} \simeq \frac{\sqrt{13200 + 1875 + 8680}}{13200} \simeq 1.17\%, \tag{20}$$

consistent with the quoted 1.19%.

For the 1 TeV benchmark, using

$$\sigma_{\mathrm{sig}} = 0.39\ \mathrm{pb}, \quad \epsilon_{\mathrm{sig}} = 0.21, \quad \sigma_{\mathrm{bgd-Z}} = 0.50\ \mathrm{pb}, \quad \epsilon_{\mathrm{bgd-Z}} = 0.025,$$

$$\sigma_{\text{bgd}-\text{mj}} = 563.8 \text{ pb}, \quad \epsilon_{\text{bgd}-\text{mj}} = 1.23 \times 10^{-4},$$

one obtains

$$\Delta\sigma/\sigma \simeq 0.49\%$$

if $L_{\text{int}} = 1000 \text{ fb}^{-1}$, but

$$\Delta\sigma/\sigma \simeq 0.69\%$$

if $L_{\text{int}} = 500 \text{ fb}^{-1}$. This is the origin of the consistency warning above.

Using the external uncertainty

$$\frac{\Delta(g_{hWW}^2/\Gamma_h)}{g_{hWW}^2/\Gamma_h} = 0.985\%,$$

the quoted coupling precisions are reproduced:

$$\frac{\Delta g_{hbb}}{g_{hbb}} = \frac{1}{2}\sqrt{(1.19\%)^2 + (0.985\%)^2} \simeq 0.77\%, \tag{21}$$

$$\frac{\Delta g_{hbb}}{g_{hbb}} = \frac{1}{2}\sqrt{(0.69\%)^2 + (0.985\%)^2} \simeq 0.60\%. \tag{22}$$

4 Broader Higgs-coupling reach: global-fit picture

4.1 The κ-framework

In the κ-framework, Higgs couplings are rescaled relative to the SM:

$$g_{hXX} = \kappa_X \, g_{hXX}^{\text{SM}}. \tag{23}$$

The relevant modifiers include:

$$\kappa_W, \ \kappa_Z : \text{couplings to weak vector bosons}, \tag{24}$$

$$\kappa_b, \ \kappa_c, \ \kappa_t, \ \kappa_\tau, \ \kappa_\mu : \text{fermion Yukawa modifiers}, \tag{25}$$

$$\kappa_g, \ \kappa_\gamma, \ \kappa_{Z\gamma} : \text{effective loop-induced modifiers}. \tag{26}$$

Additional BSM branching fractions can be included:

$$\text{Br}_{\text{inv}}, \qquad \text{Br}_{\text{und}}.$$

A measured signal strength behaves schematically as

$$\mu(i \to h \to f) \sim \frac{\kappa_i^2 \kappa_f^2}{\kappa_h^2}, \tag{27}$$

where κ_h^2 encodes the total-width rescaling. This is why a global fit is needed: individual channels measure products of production and decay couplings divided by the total width.

4.2 The $\kappa_V \leq 1$ condition

The condition

$$\kappa_V \leq 1, \qquad V = W, Z, \tag{28}$$

is often imposed in future-collider Higgs fits when the total Higgs width is not directly and model-independently measured. It helps break a flat direction between the total width and the couplings. This is a useful benchmark assumption, but it is not fully model independent; some BSM scenarios can violate it.

How to phrase this in a talk

Say: "The fit uses the standard $\kappa_V \leq 1$ benchmark assumption to control the Higgs-width flat direction. This is useful for comparisons, but it is not a fully model-independent statement."

4.3 How the LHμC projection is obtained

The global-fit result in the slide should be described as an **exploratory projection**. The idea is:

1. start from LHeC Higgs signal-strength projections;
2. keep the LHeC systematic scale factors fixed;
3. rescale statistical uncertainties using LHμC benchmark cross sections and luminosities;
4. perform the κ-framework fit using SMEFiT.

This is not yet a full detector-level LHμC Higgs global fit.

4.4 Vector, fermion, and loop modifier plots

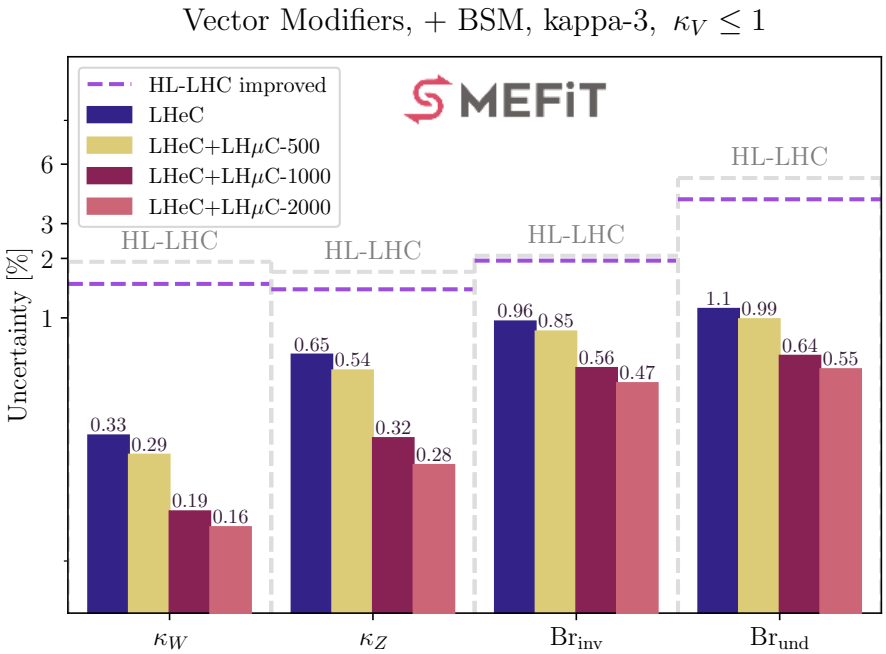
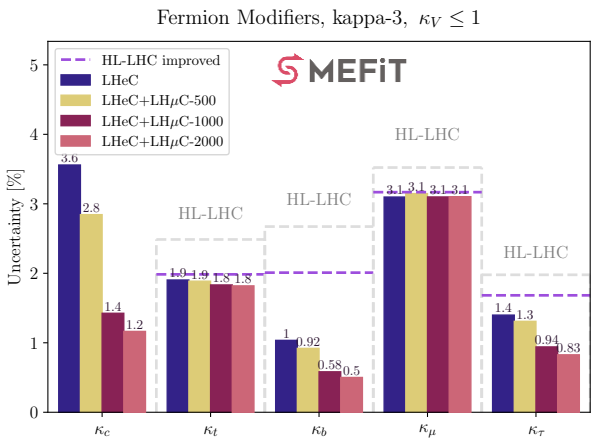
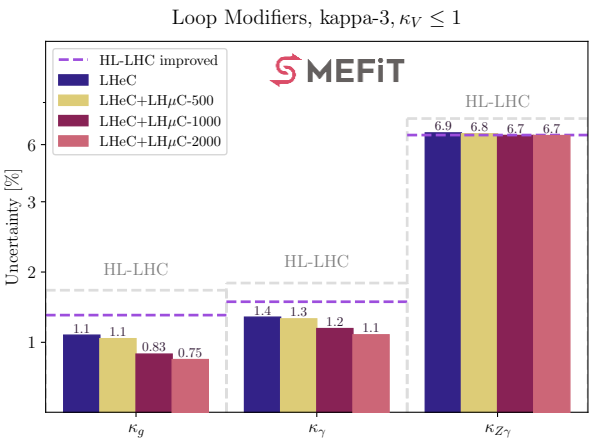


Figure 1: Vector-modifier uncertainties in the κ-3 framework. The clearest improvements appear in κ_W and κ_Z, especially for higher-energy LHμC stages.



(a) Fermion modifiers.



(b) Loop-induced modifiers.

Figure 2: Fermion and loop-induced modifier projections. In the displayed setup, the most visible LHμC improvement is in the vector sector.

The vector panel shows that κ_W improves from about 0.33% for the LHeC-only case to about 0.29%, 0.19%, and 0.16% for LHeC+LHμC-500, LHeC+LHμC-1000, and LHeC+LHμC-2000 respectively. The κ_Z uncertainty also improves. The physics reason is simple: LHμC Higgs production is mostly weak-boson fusion.

Main message of the global-fit slide

Because Higgs production at a lepton–hadron collider is driven by weak-boson fusion, LHμC most directly improves the vector Higgs couplings, especially κ_W and κ_Z. The result is promising, but should be called an exploratory projection.

5 Why LHμC is a distinctive BSM machine

5.1 The basic mechanism

The defining BSM advantage of LHμC is access to direct partonic initial states

$$\mu q, \qquad \mu g, \qquad \gamma\gamma, \qquad \gamma q, \tag{29}$$

at multi-TeV energy. The most important resonant channels are

$$\mu q \to X, \qquad \mu g \to X. \tag{30}$$

At a pp collider, many new colored particles are searched for through pair production,

$$pp \to X\bar{X},$$

or through deviations in high-mass tails. At LHμC, if a new particle couples directly to a muon and a quark or gluon, it can be produced singly. This can be a major advantage because single production has a lower threshold than pair production.

5.2 Natural BSM examples

- RPV squarks: $\mu q \to \tilde{q}$;
- leptoquarks: $\mu q \to \mathrm{LQ}$;
- color-octet muons / leptogluons: $\mu g \to \mu_8$;
- excited muons;
- contact interactions;
- photon-induced anomalous dipole and quartic-gauge-coupling probes.

Best sentence for the BSM motivation slide

LHμC is not just a higher-energy DIS machine; it is a direct $\mu q/\mu g$ resonance machine for new physics coupled to second-generation leptons.

6 RPV squarks

6.1 R-parity violation

If R-parity is violated, the MSSM superpotential may contain

$$W_{\mathrm{RPV}} = \epsilon_i L_i \cdot H_u + \frac{1}{2}\lambda_{ijk} L_i \cdot L_j \bar{E}_k + \lambda'_{ijk} L_i \cdot Q_j \bar{D}_k + \frac{1}{2}\lambda''_{ijk} \bar{U}_i \bar{D}_j \bar{D}_k. \tag{31}$$

The LQD-type term,

$$\lambda'_{ijk} L_i Q_j \bar{D}_k,$$

is directly relevant for lepton–hadron colliders because it couples a lepton, a quark, and a squark. For $i = 2$, the lepton is a muon-type lepton.

6.2 Benchmark channels

The benchmark processes can be written schematically as

$$p\mu^+ \to d\,\mu^+ \qquad \text{neutral-current-like channel,} \tag{32}$$

$$p\mu^+ \to \bar{d}\,\bar{\nu}_\mu \qquad \text{charged-current-like channel.} \tag{33}$$

The neutral-current channel can be stronger because it can involve valence-quark PDFs and contains a high-p_T muon in the final state.

6.3 How the exclusion curves are obtained

The workflow is:

1. turn on one RPV coupling at a time;
2. scan the squark mass and the coupling;
3. generate signal and background with MG5_aMC@NLO using an RPV-MSSM UFO model;
4. shower/hadronize with Pythia6;
5. cluster jets with FastJet;
6. simulate detector effects with Delphes;
7. apply event selection, including at least one jet and exactly one muon in the NC case;
8. apply a hard p_T-sum threshold,
$$p_T^{j_1} + p_T^{j_2} > p_T^{\text{cut}};$$
9. compute
$$S_{\text{stat}} = \frac{N_s}{\sqrt{N_b}};$$
10. define the 95% CL exclusion by $S_{\text{stat}} = 2$.

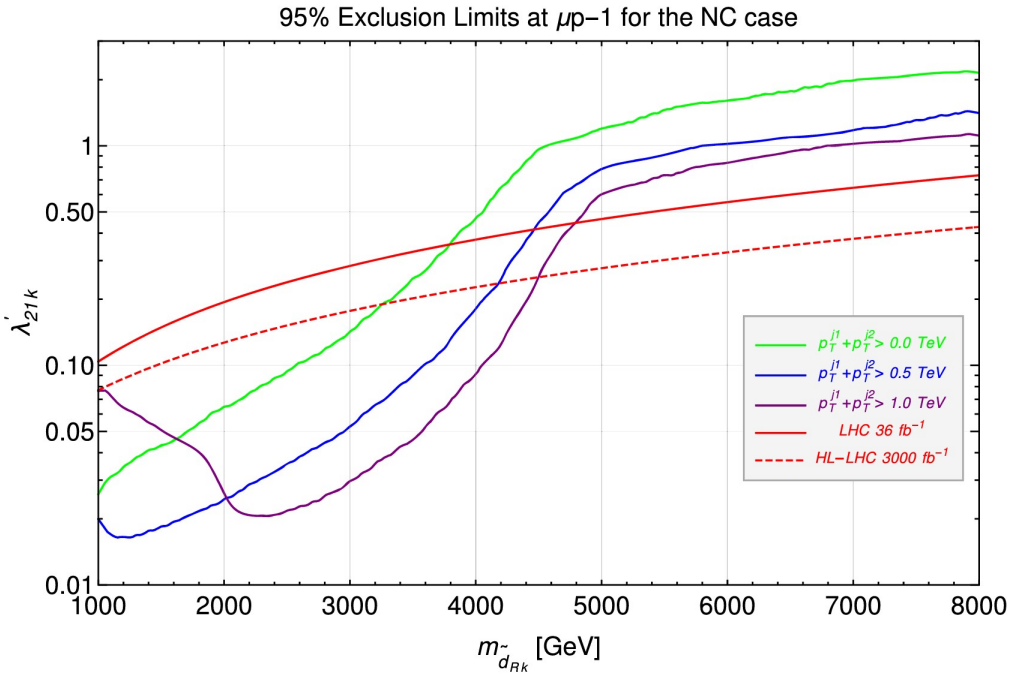


Figure 3: Representative 95% CL exclusion limits for an RPV neutral-current benchmark. The horizontal axis is the squark mass, while the vertical axis is the RPV coupling. Lower curves mean stronger sensitivity. Different colored curves correspond to different jet-p_T-sum thresholds. The red curves show LHC and HL-LHC bounds.

The green curve uses no hard p_T-sum threshold. It is strongest at low mass but weakens at high mass. The blue and purple curves impose harder cuts. They are weaker at low mass but better for heavier squarks, whose decay products are naturally harder.

RPV takeaway

In favorable neutral-current benchmark channels, LHμC can probe

$$\lambda'_{2j1} \sim 0.01\text{–}0.02$$

for squark masses around 1–3 TeV*. This can approach an order-of-magnitude improvement over LHC bounds in the on-shell region, but it is a benchmark-specific statement.*

7 Color-octet muons / leptogluons

7.1 What is μ_8?

A color-octet muon, denoted μ_8, is a hypothetical strongly interacting partner of the muon. It appears in composite-lepton scenarios, where Standard Model leptons may have more fundamental constituents. The SM muon is a color singlet, while μ_8 transforms as an octet under $SU(3)_c$.

7.2 Effective interaction

A common effective interaction is

$$\mathcal{L} = \frac{1}{2\Lambda}\left\{\bar{l}_8^\alpha g_s G_{\mu\nu}^\alpha \sigma^{\mu\nu}\left(\eta_L l_L + \eta_R l_R\right) + \text{h.c.}\right\}. \tag{34}$$

Here Λ is the compositeness scale, g_s is the strong coupling, $G_{\mu\nu}^\alpha$ is the gluon field strength, l_8^α is the color-octet lepton, and η_L, η_R are chirality coefficients. The benchmark usually assumes

$$\Lambda = m_{\mu_8}.$$

7.3 Why μp is special

At pp colliders, μ_8 searches mainly rely on pair production:

$$pp \to \mu_8 \bar{\mu}_8.$$

At a μp collider, the incoming muon can scatter directly on a gluon:

$$\mu g \to \mu_8 \to \mu g.$$

This single-resonance channel benefits from the gluon PDF, has a lower mass threshold than pair production, and gives a clean $\mu + j$ final state.

7.4 Analysis strategy

The benchmark analysis uses:

- signal: $\mu p \to \mu_8 \to \mu j$;
- background: SM $\mu p \to \mu j$;
- exactly one muon and one jet;
- $p_T^\mu > 400$ GeV, $p_T^j > 400$ GeV;
- $|\eta_j| < 3.5$, $|\eta_\mu| < 3$;
- extraction of signal and background from the $m_{\mu j}$ spectrum;
- significance

$$Z = \frac{S}{\sqrt{B}}.$$

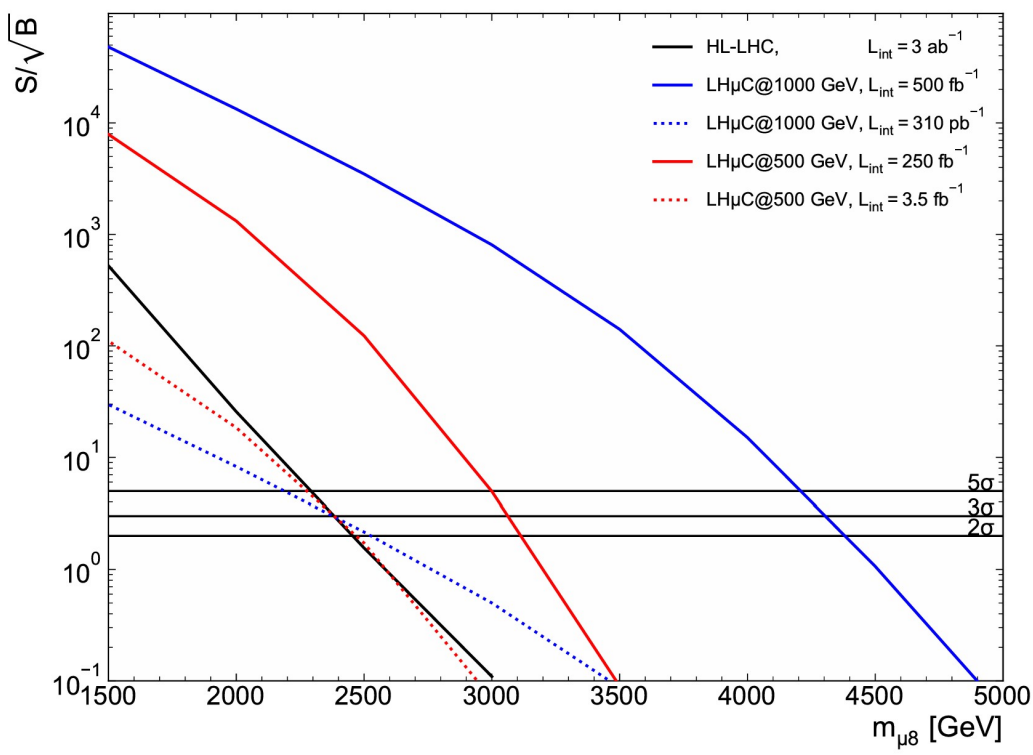


Figure 4: Representative statistical significance for color-octet muon searches. The horizontal axis is m_{μ_8}; the vertical axis is $S/\sqrt{B}$. The horizontal lines mark 2σ, 3σ, and 5σ. The LHμC curves extend to higher mass than the HL-LHC benchmark because $\mu g \to \mu_8$ is a direct single-resonance channel.

The representative 5σ reaches are approximately

$$m_{\mu_8}^{5\sigma} \sim 2.3 \text{ TeV} \quad (HL-LHC), \qquad 3.0 \text{ TeV} \quad (LH\mu C\text{-}500), \qquad 4.2 \text{ TeV} \quad (LH\mu C\text{-}1000).$$

Best slide sentence

The key advantage is the resonant μg channel,

$$\mu g \to \mu_8 \to \mu g,$$

which is absent in pp pair-production searches.

8 How to explain each plot

8.1 $h \to b\bar{b}$ table

The table summarizes signal and background cross sections and the final projected rate and coupling sensitivities. The key point is not only that the signal rate is large, but that the backgrounds can be suppressed enough by the two-b-jet requirement and the Higgs mass window to make a percent-level rate measurement plausible.

Suggested caption:

> Cut-based charged-current $h \to b\bar{b}$ benchmark. The quoted g_{hbb} reach follows from the measured $j\bar{\nu}_\mu b\bar{b}$ rate after propagating external information on g_{hWW}^2/Γ_h.

8.2 Vector κ-modifier plot

The vector plot shows projected uncertainties on κ_W, κ_Z, $\mathrm{Br}_{\mathrm{inv}}$, and $\mathrm{Br}_{\mathrm{und}}$. The strongest trend is the improvement in κ_W and κ_Z. The reason is that LHμC Higgs production is dominated by weak-boson fusion.

Suggested caption:

> Exploratory κ-framework projection using SMEFiT. LHμC improves the vector modifiers most strongly, under LHeC-like systematic assumptions and $\kappa_V \leq 1$.

8.3 RPV plot

The RPV plot shows 95% CL exclusion curves in the plane of squark mass and RPV coupling. Lower curves are better. Harder p_T-sum cuts are useful at high mass because heavy resonances produce harder jets.

Suggested caption:

> RPV squark coupling reach from μq-initiated resonance production. Different curves show different jet-p_T-sum thresholds; lower curves correspond to stronger coupling reach.

8.4 μ_8 plot

The μ_8 plot shows $S/\sqrt{B}$ as a function of m_{μ_8}. The LHμC curves remain above the 5σ line to larger masses than the HL-LHC benchmark. This is the visual demonstration of the single-resonance μg advantage.

Suggested caption:

> Leptogluon reach from resonant $\mu g \to \mu_8 \to \mu g$. The LH$\mu$C curves extend to higher masses than the HL-LHC pair-production benchmark.

9 Improved slide text

9.1 Slide: concrete Higgs benchmark

Improved title: Concrete Higgs benchmark: $h \to b\bar{b}$ in charged-current VBF

Improved bullets:

- Benchmark channel:
$$p\mu^+ \to j\bar{\nu}_\mu h, \qquad h \to b\bar{b}.$$
- Simulation chain: MG5_aMC@NLO + Pythia6 + FastJet + Delphes.
- LHeC-like asymmetric detector setup with 75% b-tagging.
- Selection: exactly two b-jets and $|m_{bb} - m_h| < 25$ GeV.
- Rate interpretation:
$$\sigma(j\bar{\nu}_\mu b\bar{b}) \propto \frac{g_{hWW}^2 g_{hbb}^2}{\Gamma_h}.$$

Key finding: Already with LHeC-like detector assumptions, LHμC reaches a sub-percent g_{hbb} benchmark sensitivity. Final robustness requires validated luminosity, detector/MDI background, and systematic assumptions.

9.2 Slide: global Higgs fit

Improved title: Exploratory global Higgs fit: vector modifiers benefit most

Improved bullets:

- Beyond the dedicated $h \to b\bar{b}$ benchmark, we perform an exploratory κ-framework fit.
- LHeC systematic scale factors are kept fixed.
- Statistical uncertainties are rescaled using LHμC benchmark Higgs rates.
- The fit is performed with SMEFiT.
- The clearest projected improvement is in the vector sector:
$$\delta\kappa_W \simeq 0.29\% \ (LH\mu C\text{-}500), \qquad 0.19\% \ (LH\mu C\text{-}1000).$$

Caveat: The fit assumes $\kappa_V \leq 1$ to control the Higgs-width flat direction; this is a standard benchmark assumption, not a fully model-independent statement.

9.3 Slide: why LHμC is a BSM machine

Improved title: Why LHμC is a distinctive BSM machine

Improved bullets:

- LHμC combines a point-like muon beam with quark and gluon partons at multi-TeV energy.
- Its channel-specific advantage is direct access to
$$\mu q \to X, \qquad \mu g \to X.$$
- This enables resonant single production of new states coupled to second-generation leptons.
- Representative portals: RPV squarks, leptoquarks, color-octet muons, excited leptons, and contact interactions.
- Compared with pp, LHμC can test selected models through single-resonance production rather than only pair production or inclusive tails.

9.4 Slide: representative BSM benchmarks

Improved title: Benchmark BSM portals: μq squarks and μg leptogluons

Improved bullets:

- **RPV SUSY:** LQD-type couplings give resonant squark production in μq scattering. In favorable NC channels,
$$\lambda'_{2j1} \sim 0.01{-}0.02$$
can be probed for $m_{\tilde{q}} \sim 1{-}3$ TeV.
- **Color-octet muon / leptogluon:**
$$\mu g \to \mu_8 \to \mu g.$$
This direct μg resonance is absent in pp pair-production searches.
- Representative 5σ reach:
$$HL-LHC: \ 2.3 \text{ TeV}, \qquad LH\mu C\text{-}500: \ 3.0 \text{ TeV}, \qquad LH\mu C\text{-}1000: \ 4.2 \text{ TeV}.$$

Caveat: Representative benchmark studies; final reach requires detector/MDI backgrounds, systematic uncertainties, and channel-specific optimization.

10 Speaker notes

10.1 $h \to b\bar{b}$

This is the most concrete Higgs benchmark in the talk. The process is charged-current VBF, $p\mu^+ \to j\bar{\nu}_\mu h$, followed by the dominant Higgs decay $h \to b\bar{b}$. The signal is selected with two b-jets and a Higgs mass window. The measured rate scales as $g^2_{hWW} g^2_{hbb}/\Gamma_h$, so after using external information on g^2_{hWW}/Γ_h, we obtain a projected g_{hbb} sensitivity. The key point is that the large LHμC rate turns this into a sub-percent benchmark, although the final number depends on luminosity and systematic assumptions.

10.2 Global fit

The previous slide was a single-channel study. This slide asks what happens in a broader Higgs coupling interpretation. We use a κ-framework fit, implemented with SMEFiT. This is exploratory: we keep the LHeC systematic scale factors and rescale the statistical part using the LHμC rates. The clearest improvement is in the vector modifiers, especially κ_W and κ_Z, which is natural because Higgs production at a lepton–hadron collider is primarily weak-boson fusion.

10.3 BSM motivation

The BSM point is simple but important. In pp, many colored new particles are searched for through pair production, which costs energy because two heavy particles must be produced. At LHμC, if the new state couples to a muon and a quark or gluon, it can be produced singly as a resonance. This is why RPV squarks and color-octet muons are natural benchmark examples.

10.4 BSM benchmarks

These two examples show the same BSM principle in two different initial states. RPV squarks use μq resonance production through LQD-type couplings, and favorable NC channels can probe λ' around 10^{-2}. Color-octet muons use the μg initial state. The single-resonance channel $\mu g \to \mu_8 \to \mu g$ gives a stronger reach than HL-LHC pair production in this benchmark. The exact numbers should be understood as representative, pending full detector and MDI studies.

11 Likely audience questions and compact answers

Q: How robust is the hbb sensitivity once systematics are included? The quoted rate precision is primarily statistical. The coupling extraction includes external information on g^2_{hWW}/Γ_h, but a final study must include b-tagging, jet-energy scale, background normalization, theory/PDF uncertainties, and beam-induced backgrounds.

Q: Why do you need external information on g^2_{hWW}/Γ_h? Because the measured rate is proportional to $g^2_{hWW}g^2_{hbb}/\Gamma_h$. A single rate cannot isolate g_{hbb} without information on the production coupling and total width combination.

Q: Is the global fit independent, or mostly a rescaling? It should be presented as an exploratory projection. Statistical uncertainties are rescaled using LHμC cross sections, while LHeC systematic factors are retained.

Q: Why impose $\kappa_V \leq 1$? It controls the Higgs-width flat direction in fits without a direct model-independent Higgs width measurement. It is useful for comparison, but not fully model independent.

Q: Why are RPV squarks like leptoquarks here? The LQD operator couples a lepton, a quark, and a squark. Phenomenologically, the squark can appear as a leptoquark-like resonance in μq scattering.

Q: Are interference effects included in the RPV benchmark? The benchmark treatment neglects subdominant interference between signal and background. A final study should include it.

Q: Why is μp better than pp for μ_8? Because μp provides the direct single-resonance process $\mu g \to \mu_8$, whereas pp searches mainly rely on pair production.

Q: How model-dependent is the μ_8 reach? It depends strongly on the effective operator, the chirality assumptions, and especially Λ. The benchmark assumes $\Lambda = m_{\mu_8}$.

Q: What about muon-decay beam-induced backgrounds? This is a central detector/MDI issue. The benchmark studies are promising, but realistic shielding, timing, and detector-response simulations are needed before final experimental claims.

12 Final checklist before the presentation

Must fix

1. Resolve the hbb luminosity consistency issue for the 1 TeV benchmark.
2. Label σ_{sig} clearly: inclusive h production or final $b\bar{b}$ state.
3. Replace “explorative” with “exploratory”.
4. Add one caveat line to the global-fit slide: LHeC-like systematic factors are retained.
5. Fix spacing: “LHμC benchmark”, not “LHμCbenchmark”.

Should improve if time

1. Add a short explanation of $\kappa_V \leq 1$.
2. Mention $\Lambda = m_{\mu_8}$ in the μ_8 caption or speaker note.
3. Make plot captions carry the physics message.
4. Replace "Color-octet muon μ_8 reach" by "Leptogluon reach from resonant $\mu g \to \mu_8 \to \mu g$".

13 Further reading map

Table 2: Suggested references and what to learn from each.

Topic	Reference	Why read it
LHμC concept and HELP4CERN physics case	HELP4CERN white paper draft	Overall collider motivation, benchmark cross sections, Higgs, $\gamma\gamma$, top, and BSM overview.
Muon–proton Higgs and RPV benchmarks	Cheung and Wang, Phys. Rev. D 103, 116009 (2021), arXiv:2101.10476	Original μp benchmark for hbb and RPV squark reach.
Future Higgs κ-framework	de Blas et al., JHEP 01 (2020) 139, arXiv:1905.03764	Standard future-collider Higgs comparison and κ-framework context.
Future collider SMEFT fits	de Blas et al., arXiv:2206.08326	SMEFT global-fit context and future-collider projections.
SMEFiT and precision reach	Armadillo et al., arXiv:2604.16596	Recent multi-framework precision interpretation and updated SMEFiT release.
μLHC and color-octet muons	Akturk et al., arXiv:2506.15445	μ^+p collider design and color-octet muon benchmark.
Historical RPV at μp	Carena, Choudhury, Quigg, Raychaudhuri, Phys. Rev. D 62, 095010	Classic study of RPV resonances at muon–proton colliders.
Leptogluon phenomenology	Goncalves-Netto et al.; Acar–Kaya–Oner	Color-octet lepton / leptogluon theory and collider phenomenology.

References

[1] D. Acosta et al., *White Paper on High Energy, High Luminosity Lepton–Hadron Colliders at CERN*, HELP4CERN Study Group draft.

[2] H. Khanpour, *Perspectives for HET physics studies at a muon-hadron collider at CERN*, DIS 2026 presentation draft.

[3] K. Cheung and Z. S. Wang, *Physics potential of a muon-proton collider*, Phys. Rev. D **103**, 116009 (2021), doi:10.1103/PhysRevD.103.116009, arXiv:2101.10476.

[4] J. de Blas et al., *Higgs Boson Studies at Future Particle Colliders*, JHEP **01** (2020) 139, arXiv:1905.03764.

[5] J. de Blas, Y. Du, C. Grojean, J. Gu, V. Miralles, M. E. Peskin, J. Tian, M. Vos and E. Vryonidou, *Global SMEFT Fits at Future Colliders*, Snowmass 2021, arXiv:2206.08326.

[6] T. Armadillo, E. Celada, J. ter Hoeve, F. Maltoni, L. Mantani, J. Rojo, A. N. Rossia, S. Tentori, M. O. A. Thomas and E. Vryonidou, *New Physics Reach through Precision at Future Colliders: a Multi-Pronged Approach*, arXiv:2604.16596.

[7] D. Akturk et al., *μLHC: Antimuon Ring and HL-LHC based μ^+p Collider*, arXiv:2506.15445.

[8] M. Carena, D. Choudhury, C. Quigg and S. Raychaudhuri, *Study of R-parity violation at a muon proton collider*, Phys. Rev. D **62**, 095010 (2000), arXiv:hep-ph/0006144.

[9] D. Goncalves-Netto, D. Lopez-Val, K. Mawatari, I. Wigmore and T. Plehn, *Looking for leptogluons*, Phys. Rev. D **87** (2013).

[10] Y. C. Acar, U. Kaya and B. B. Oner, *Resonant production of color octet muons at Future Circular Collider-based muon-proton colliders*, Chin. Phys. C **42** (2018) 083108.

[11] J. Alwall et al., *MadGraph 5: Going Beyond*, JHEP **06** (2011) 128.

[12] M. Cacciari, G. P. Salam and G. Soyez, *FastJet user manual*, Eur. Phys. J. C **72** (2012) 1896.

[13] J. de Favereau et al., *DELPHES 3: a modular framework for fast simulation of a generic collider experiment*, JHEP **02** (2014) 057.